

University of British Columbia, Vancouver

Department of Computer Science

Milestone #: 3

Date: March 11, 2025

Group Number: Group 30

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By typing our names and student numbers in the above table, we certify that the work in the attached assignment was performed solely by those whose names and student IDs are included above. (In the case of Project Milestone 0, the main purpose of this page is for you to let us know your e-mail address, and then let us assign you to a TA for your project supervisor.)

In addition, we indicate that we are fully aware of the rules and consequences of plagiarism, as set forth by the Department of Computer Science and the University of British Columbia

CPSC 304 Project Timeline

Team Name: Team 30

Team Members: Hongkai Liu, Cillian Dong, Jason Ji

Date Created: March 11, 2025

Project Deliverables & Deadlines

Milestone	Deadline
Milestone 4 (Implementation)	April 2, 10:00 PM
Milestone 6 (Peer and Self-Evaluation)	April 2, 10:00 PM
Milestone 5 (Demo Presentation)	April 7, 11:59 PM

Phase 1: Database Setup & GUI Design (March 17 - March 23)

Objective:

Ensure database schema is finalized and the GUI structure is in place.

Tasks:

1. Review and finalize ER diagram & schema. (ALL)
2. Implement SQL DDL, including CREATE TABLE, foreign keys, constraints. (Cillian)
3. Insert example data into the database. (Cillian)
4. Write SQL script for dropping & recreating tables. (Hongkai)
5. Establish database connection from backend. (Hongkai)
6. Design GUI wireframes (layouts for forms, tables, buttons). (Jason)
7. Setup frontend framework (HTML + CSS + JavaScript(Node.js) / React). (Jason)

Challenges & Remaining Work:

- Ensuring database constraints work as expected.
- Deciding on GUI frameworks and making sure it integrates with backend queries.

Phase 2: SQL Query Implementation & GUI Integration (March 24 - March 30)

Objective:

Implement all required SQL queries and connect them to the GUI.

Tasks:

1. Implement INSERT query with GUI form. (Hongkai)
2. Implement UPDATE query with GUI. (Hongkai)
3. Implement DELETE query with GUI. (Hongkai)
4. Implement Selection query (search/filter). (Hongkai)
5. Implement Projection query (choose displayed attributes). (Jason)
6. Implement Join query (multi-table search). (Jason)
7. Implement Aggregation with GROUP BY query. (Jason)
8. Implement Aggregation with a HAVING query. (Cillian)
9. Implement Nested Aggregation with GROUP BY. (Cillian)
10. Implement Division query. (Cillian)
11. Test and debug all queries. (ALL)

Challenges & Remaining Work:

- Making sure user inputs don't cause SQL injection vulnerabilities.
- Ensuring queries return meaningful results with real data.
- Handling UI design for query input.

Phase 3: Final Testing & Milestone 4 Submission (March 31 - April 2)

Objective:

Ensure the system is stable and submit Milestone 4.

Tasks:

1. Finalize GUI layout and fix design bugs. (Jason)
2. Conduct end-to-end testing of queries with GUI. (Hongkai)
3. Write Milestone 4 Report. (ALL)
4. Submit Milestone 4 to Canvas. (Cillian)
5. Complete Milestone 6 Peer & Self Evaluation. (ALL)

Challenges & Remaining Work:

- Ensuring smooth query execution within the GUI.
- Making the user interface intuitive for non-technical users.
- Testing edge cases for database operations.

Phase 4: Milestone 5 Preparation & Final Demo (April 3 - April 7)

Objective:

Prepare and present the demo.

Tasks:

1. Re-run database setup script and test. (ALL)
2. Review and optimize SQL query performance. (ALL)
3. Final GUI improvements based on feedback. (ALL)
4. Conduct mock demo rehearsal. (ALL)
5. Present Milestone 5 Demo to TA. (ALL)

Challenges & Remaining Work:

- Avoiding runtime errors during the live demo.
- Preparing for possible TA questions on implementation.
- Making sure the demo fits within 30 minutes.